

Bioinformatics for engineers

Biological Data Bases

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Universidad Javeriana



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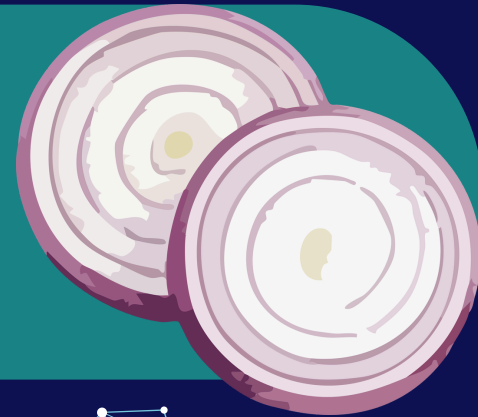
1. Thanksgiving



2. Do you remember what we mean by meta?



4. The layers



3. The growth



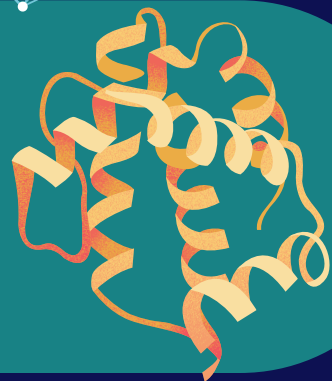
3. Databases



4. Nucleotides bases



5. Protein bases



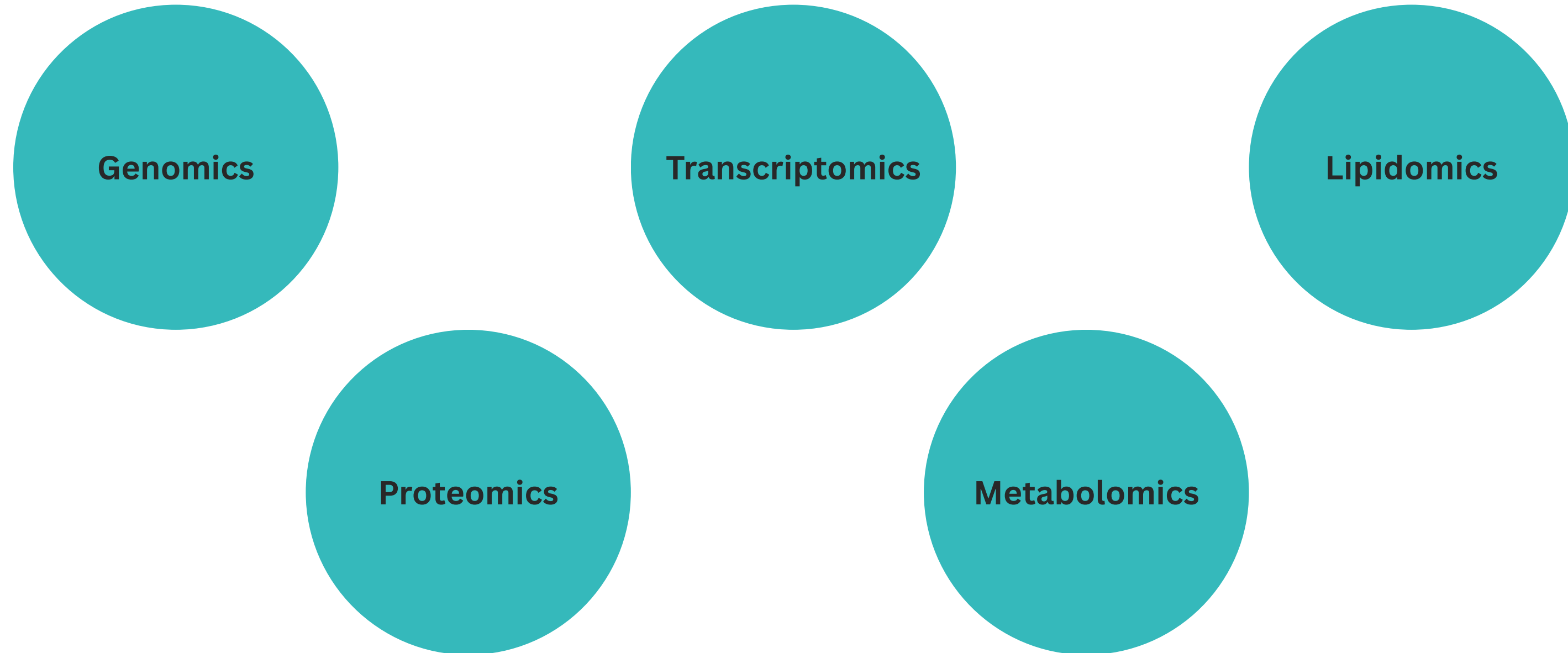
6. Specialized



THANKSGIVING

The -omics science

How to obtain the data

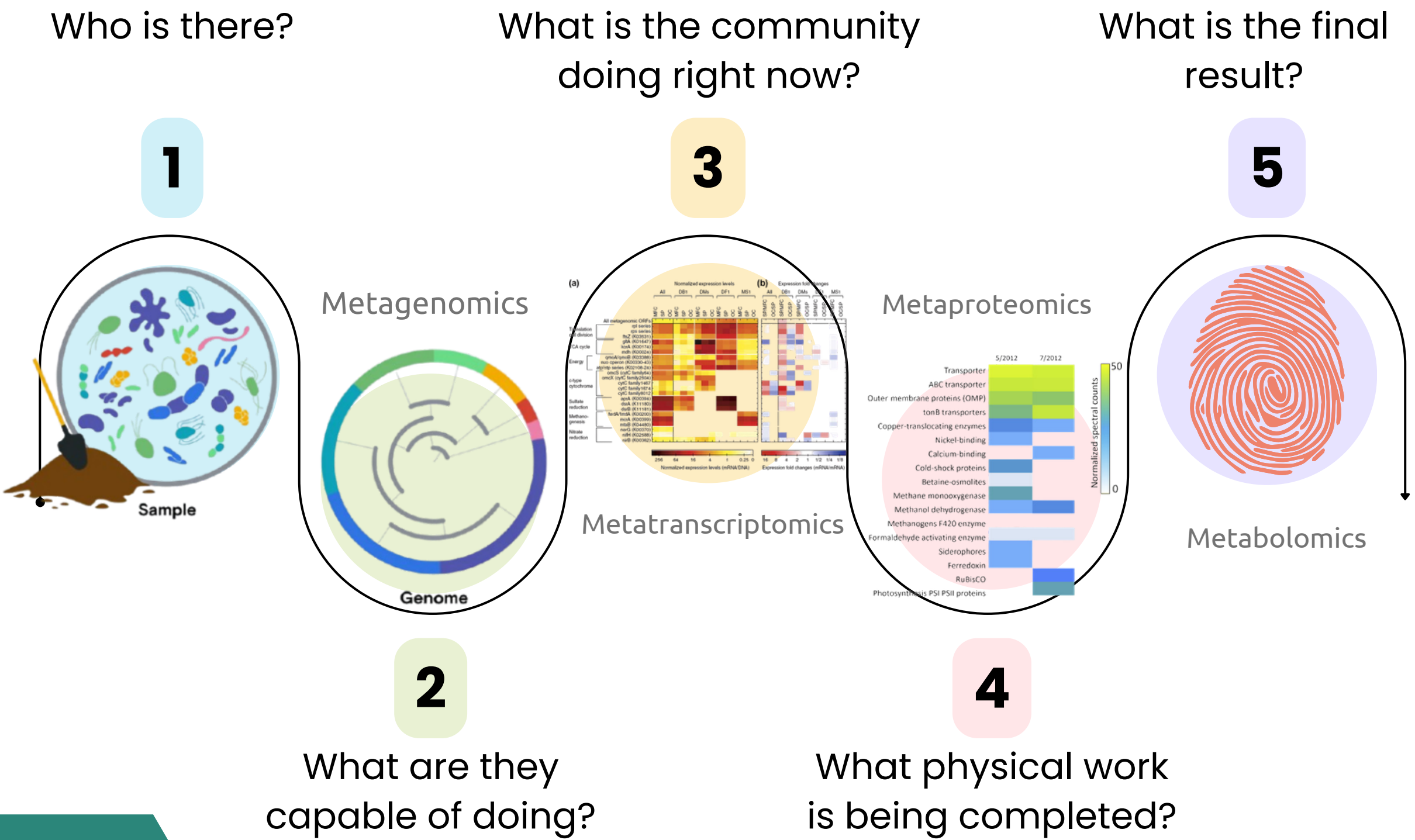


- We can use different types of data.
- Each source of information requires multiple type of experimental process to obtain the data.
- 1. • The high cost associated with this techniques stimulates the international collaborations..

THROW BACK TUESDAY!!!

The meta/horizontal view

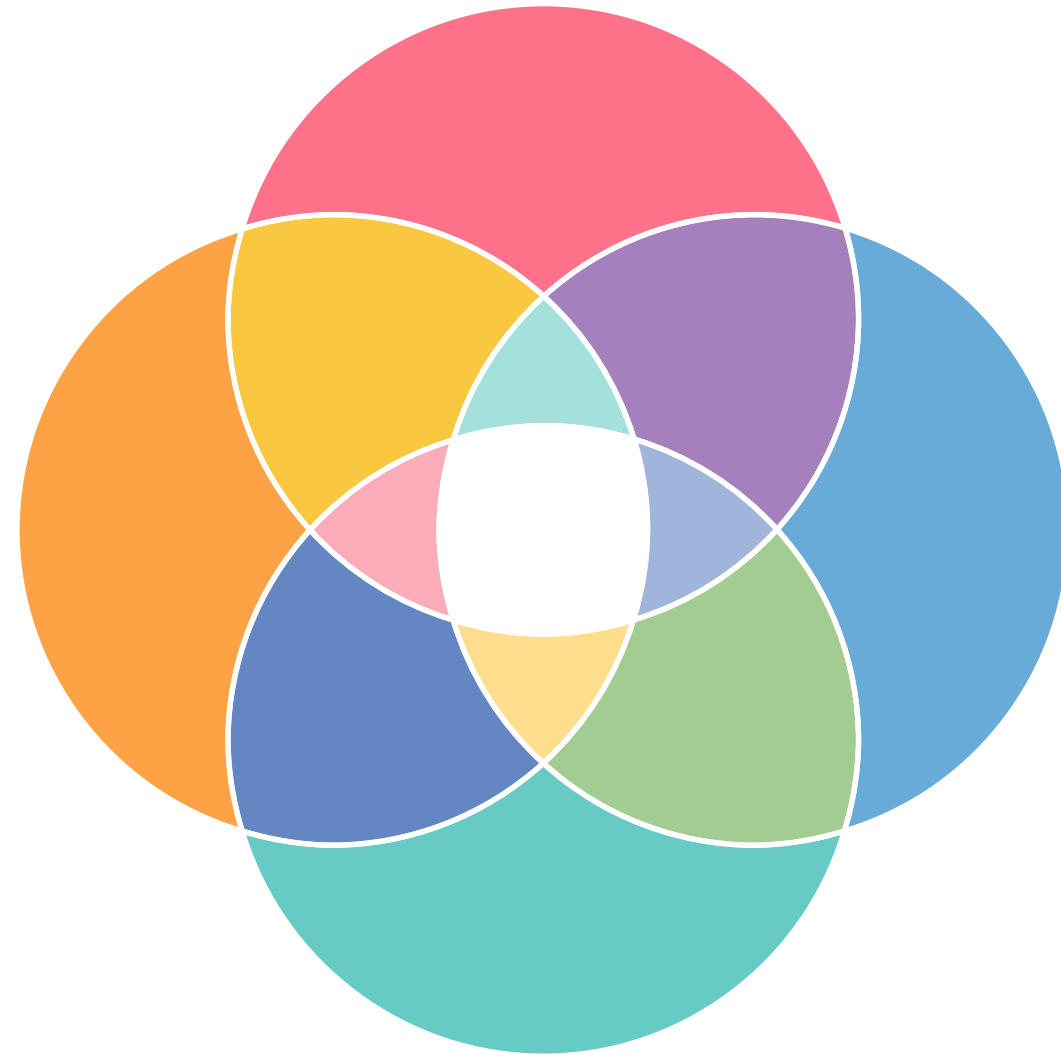
We aim to understand whole communities of organisms, not only one individual.



**SO WHAT IS META
ANALYSIS?**

An investigation philosophy

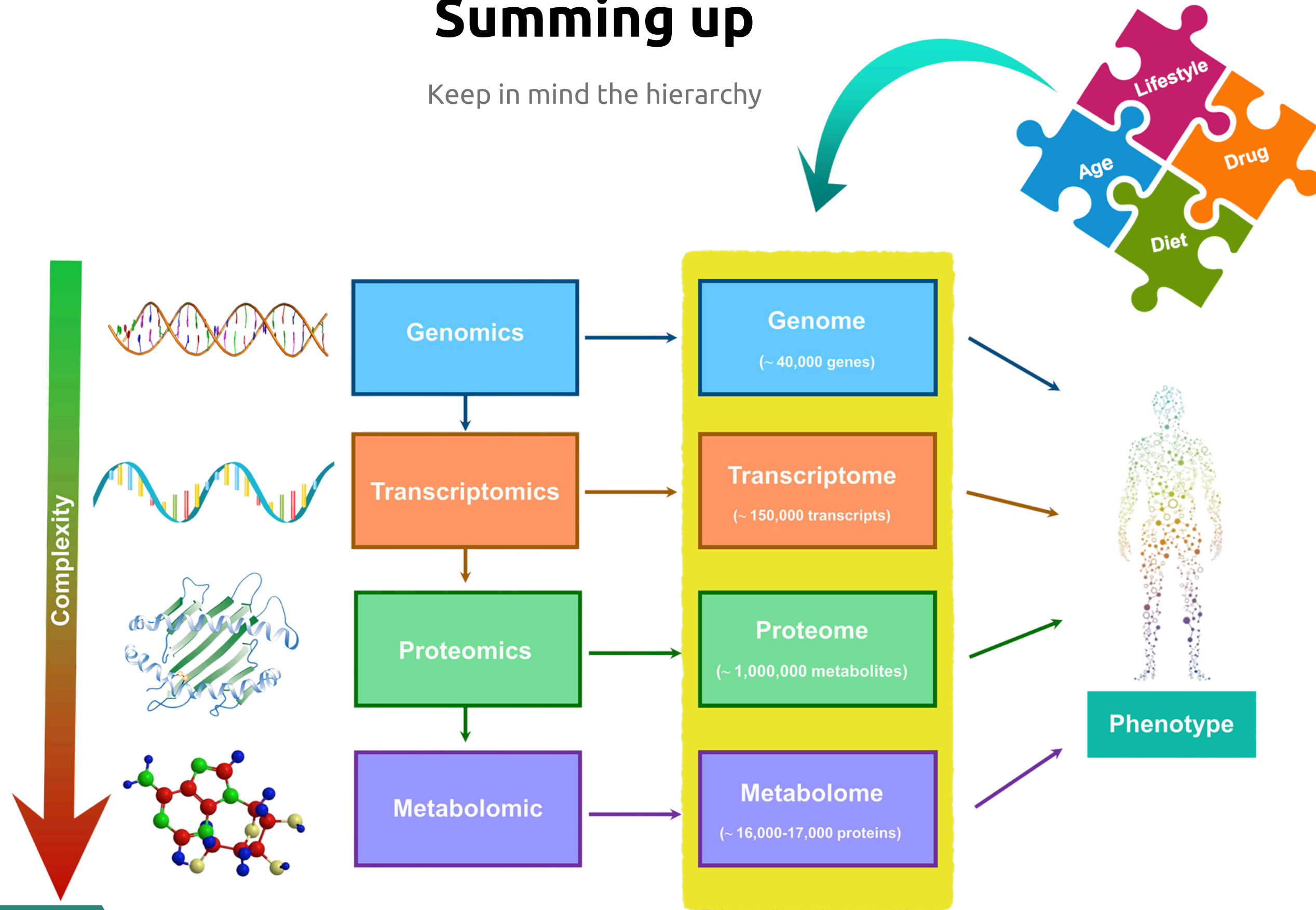
An integrated study



- We expect to find information from different organisms, from a **whole living community**.
- Thus, we are not obtaining the organized puzzle.
- The big deal is achieving the statistical and computational integration of these datasets in order to make conclusions with statistical value.

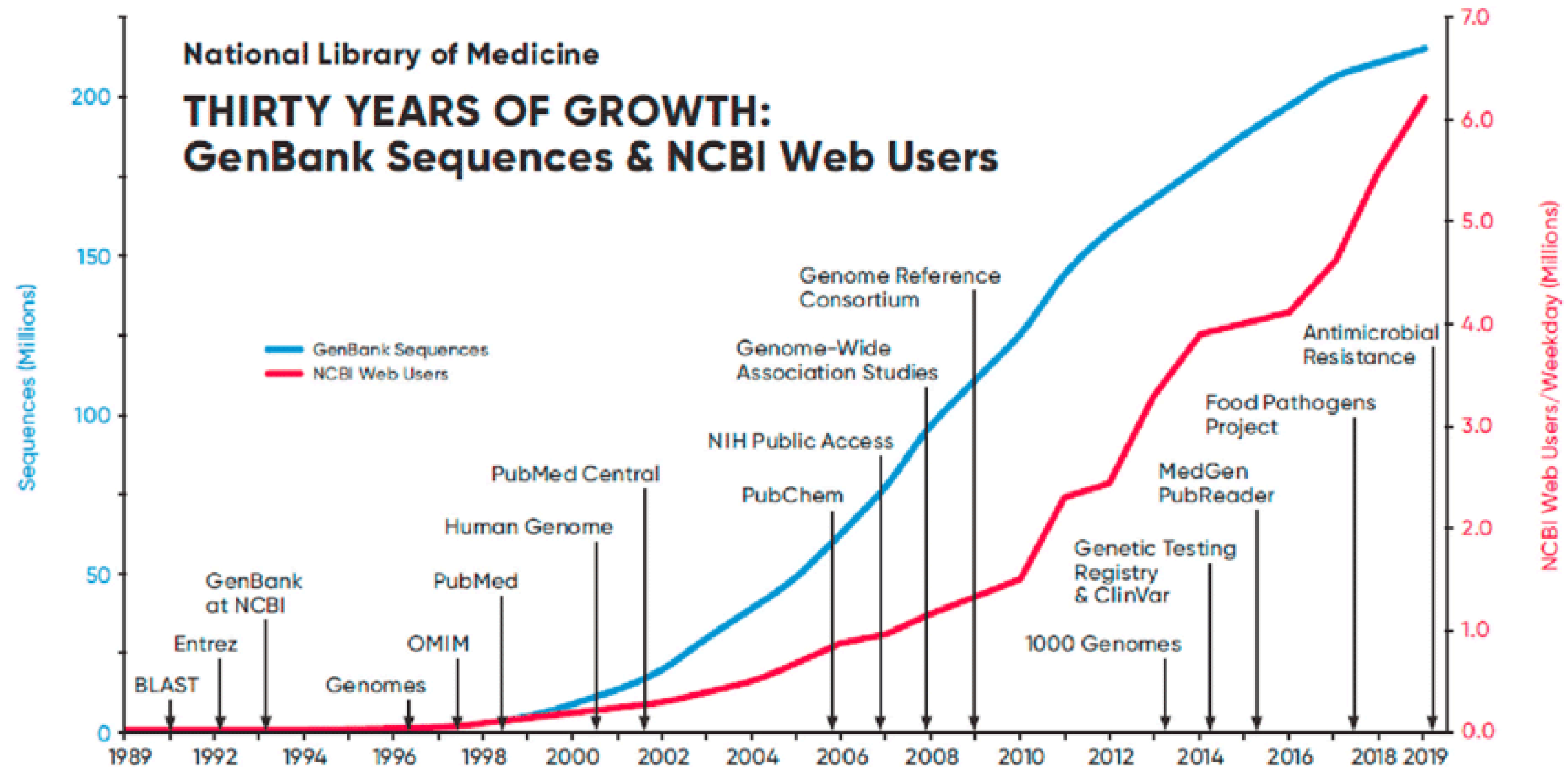
Summing up

Keep in mind the hierarchy



THE GROWTH

How much information can we find?



APPRIS, AREK, ASDb, ArcBase, circCircNet, CircNet, Cistrome Data Browser, EndoQuad, eRNABase, EVLncRNA, exorBases, Expression Atlas, Expression Atlas, Factorbook, FL-cirAtlas, FliBase, G4LD, GEO, HNCDB, HOMOCOMM, IMG/PR, JASPAR, JASPAR, KnockKT, LiRbase, LncACtDb, lncRNAsNP2, LncRNAWiki, m5CAT, atlas, m6A-Atlas, m7GHub, MAJIQforLife, MethMotif, MINIRbase, MirGeneDB, miRNAntisSenseAtlas, mirTarBase, miRTarBase, MODOMICS, MODOMICS, NAKED, NAKed, Nanobase.org, ONQUADRO, piCdb, piRBase, PLSDb, Pol3Base, POSTAR, PRODRIC, R-loopsBase, RegulonDB, ReMap, Ribosome-switch, RMBASE, RNALint, RNALocate, RNAPISeq, RSC, scaPAtLAS, sCAPAd, sCATAC-Reg, Enhancer, scGRNbase, scMetHap, SRA, Tarbase, TcofBase, Telobase, TETTS, TF-Marker, TFcheckpoint, TransLinc, LRIFtarget, tsRFun, UTEXas Aptamer Database, VirBioDB, International Nucleotide Sequence Database Collaboration, BioSample, DDBJ, EBI patent sequences, European Genome Phenome Archive (EGA), European Neuroscience Archive, FusionDB, GeneBank, NCBI BioSample/BioProject, SIS resources, The DNA Record Archive (SRA), Ubiquowser, Coding and non-coding RNA, ACLAME - A Classification of Mobile genetic Elements, ARSente, CEGA, ChITraRS, Ciliate IES-MDS database, CORG - A database for Comparative Regulatory Genomics, Dfam, DiProDB, Doric, ECRBase, Entrez Gene, etl1b, GBshare, Genetic Codes, GISD, CyDB, InStAdb, Isfinder, iL1Base, MethKBank, MethBank, MethSMRT, NcsmethDB, NPDR - Non-Genomic Positioning Region Database, OGRDB, OriDB - The DNA Replication Origin Database, PANIDT, Plant repeat databases, PolyMiTs, PseudoGene, R-loopDB, RefSeq (microbial), RegTransBase, Retroyza, S/MART DB, SINEBASE, SNPSTR, Synthetic Gene Database, TassDB, TIGR Plant Transcript Assembly Database, TranspoGene, TRDB, UCbase and miRFun, UCNBase, UgMicroSatdb, UniVec, UTUTB/UTRITS, UTRome, VectorDB, VISTA Enhancer Browser, Gene structure, introns and exons, splice sites, APPRIAS, APPRIAS, ASCiDB, ChimerDB, DBASSV3, iEcGene, EDAS - EST-Derived Alternative Splicing Database, ExTraST, HEXEVEN, Hollywood, JuncDB, MultiSpliceDB, ProSAs, REACT, SpliceAiD-F, Spleinfo, spliceNet, Spliceosome Database, UI2DB, uORFdb, uORFdb, Yeast Intronic Database, Transcriptional regulator sites and transcription factors, 3d-Footprint, ABS, ACTIVITY, AGRIIS - Arabidopsis Gene Regulatory Information System, AnimalTFDB, ASPD, BloodChip, ChiPBase, cisRED, CMCSDB, CollecT, CorneRegNet, COPRESDB, CTCF Binding Site Database, DBD, dBSUPER, DBTSS, DBTSS/DBKERO, DOOP - Databases of Orthologous Promoters, DRPP, ECRBase, Factorbook, FlyFactorSurvey, FlyTF, GeneNet, GenomeTFAdb, Greiglit, TrdB, HOMOCOMM, HOMOCOMM, HTPEXEL, JASPAR, JASPAR, MachiBase, MAPPER, ODB - Operon database, OnTheFly, oTFD, ORegAnno, PAZAR, Plant Stress-Responsive Gene Catalog, PlantProm, PreMed, PRODRIC, REDfly, RegPrecise, ScerTF, SCPD - Saccharomyces cerevisiae promoter database, SEA, SELEXdb, SKY/FISH and CGH, SNP2TFBS, SpatialDB: a database for spatially resolved transcripts, STIFDB2, SwissRegulon, Tcof-DB, Telomerase database, TESSE, TFBScan, TFBShape, TFClass, TFClass, TiProD, TRACTOR db, TRANSCOMPdb, TRANSFACdb, TransmiR, TRANSPATHdb, TREd - Transcriptional Regulatory Element Database, TSDB, TTSMI, UniPROBE, VISTA Enhancer Browser, WebGetServer, YEASTRACT, YEASTRACT, YTFaCoSb, RNA sequence databases, 16S and 23S Ribosomal RNA Mutation Database, 3D RNA modification maps, 5S Ribosomal RNA Database, AlphaFold Protein Structure Database, APASdb, AREB, BPS, BESr, Cerebral Small RNA Database, CircNet, CORG - A database for Comparative Regulatory Genomics, CoVPDB, CPLM, DARNED, DASHR, Database for Bacterial Group II Introns, dbAMP, dbPTM, dBRES, DIANA-LncBase, DisProt, doRNA, DRAMP, ELM, European rRNA database, FlyRNAi, fRNAdb, FuzDB, G protein research platform, GreenGenes, GR50B, GrDB - Genomic tRNA Database, HMMD v2.0 - Human microRNA Disease Database version 2.0, Human Proteome Atlas, iRESite, ITS2, LincSNP, LINCediting, LINCpedia, lncRNAb, lncRNAsNP, LncRNAWiki, MeSAdb, MeT-DB, MeT-DB, microRNAdb, miR2iSense, miRbase, miRDB, miRecords, miREx, miRGator, miRNAmap, miRNest, miRHost, miRPathDB, miRTarBase, MODOMICS, MODOMICS, NAPP, NATSDB, NCIR - Non-Canonical Interactions in RNA, ncRNAs Database, NONCODEv5, NPinter, NRED, PDbe-Kb, piRNABase, piRNAlocusDB, Plant snRNA DB, PlanRNA, PMIRkb, PolyASte 2.0: a consolidated atlas of polyadenylation sites from 3kÅ - 4' end sequencing, PolyAdb, Met-PolyA, Db, POSTAR, PRIDE, RADAR, RAD, REDIPortal, RefSeq, RFam, Rifam, Ribosomal Database Project (RDP-II), RiBase, RMBase, RNA Modification Database, RNA STRAND, NCANter, NCANter, RNAdb, RNAI code, RNA Junction, RNALocate, RPFdb, SCOPE, SELEXdb, siRecords, siRNAdb, snoScRNAbase, sNOPy, snoRNA-LBME-db, Somamir, sRNAmap, SRPDB, StarBase, SYNBIB, T-psi-ci user friendly database of tRNA sequences and structures, TarBase, The Small Subunit rRNA Modification Database, tmRDB, tRFdb, tRNADB-C, UTRdb/UTRsite, Vir-Mir db, ViRoBase, ViSiRNAdb, Yeast snRNA Database, YM500, Coding and non-coding RNA, EVLncRNAs, Protein sequence databases, AD3D Model, EnzymeProtein Database, AFDB, ASD, ATLAS, BioExcel-CV19, BiOLIP, ChannelsDB, ConsensusPathDB, dPAris, DesciProt, ELIN, EMDE, Equilibrator, G-proteinsDB, Gramene, GRAND, HMDB, HumanNet, IID, Inclusive, Kincore, MesSPUBet, MultifacetedProtdb, NMPIamsDB, nRD, OffactionBase, OA, Orthology, OpenProt, OrithoMAM, PeD, PLabDB, Reactome, Rhea, SingPro, SMART, SPOT, TissueNexus, UniTM, General sequence databases, CharProtDB, COMBREX, EukProt, NCBI Protein Database, NRinch, PA-GOSUB, Bio-Protein Information Database, drPF, TCDB, UnclustProt, UniPar, UniProt, UniRef, UniSave, Protein properties, AAindex, BindingDB, Cybasen, dBPDM, eProS, HHMD, iProLiNK, MALISAM, MegaMotifBase, MemMoRF, Minimoft motif, MobIDb, MP-PDF, PFD - Protein Folding Database, PIDDT, PINN, PPD, PPT-DB, Proteome-p1-2.0, ProTherm, PTARD, REFOLD, SwissSidechain, TOPPR, Protein localization and targeting, CentrosomeDB, eSLDB - eukaryotic Subcellular Localization database, Interfil, LOCATE, LoCdb, MatrixDB, Membraneome, MicroKits, NESbase, NL5db, NMPDB - Nuclear matrix associated proteins functional annotation database, NucProtein Protein Database, NPD - Nuclear Protein Database, NURSA, OGRe - Organelle Gene Retrieval, PeroxisomeDB, PSORTdb, Secreted Protein Database, SUBA, THGS - Transmembrane Helices in Genome Sequences, TMPad, TopDB, ValidNES, Protein sequence motifs and active sites, ASC - Active Sequence Collection, COME - Co-Ordination of Metals etc., CoPS, eBLOCKs, eSite - Electrostatic surface of Functional site, ELM - Eukaryotic Linear Motif: functional sites in eukaryotic proteins, FireDB, HRAp - Database of occurrence of HomoRepeats and Patterns in proteomes, InterPro, MeMotif, MetalPdb, MimeoDB, Minifit, Minifit, mutLB5geneDB, NCIO, O-GLYCBase, PDBSITE, PHOSIDA, Phospho3D, PhosphoSitesPlus, PolyQ, PRIDB, PRINTS, ProRuLe, PROSITE, ProTeus, SitEx, SuperSite, VDjDb, Protein domain databases; protein classification, 3Dspace: Description of Proteins involved in 3D domain Swapping, ADDA - A Domain database, BALiBASE, Benchmark, BIOZOON, CDD, DomIns - Database of Domain Insertions, eggNOG, EVEREST - Evolutionary Ensembles of Recurrent Segments, FunShifit, His, Inparanoid, IDom, iPaRM, iProClass, LenVarDb, MobIDb, MULPSMM, OMIA, OPTIC, OrthoDB, OrthologueDB, PALI, PANDIT, PANTHER, PDBe, Pfam, PhyloFactors, PIRSF, ProDom, Protein Clusters, ProtoNet, RBP-VAr, RBPDF, SBASE, SIMAP, SISYPHUS, SMART, sORFs, SUPFAM, TIGRFAMS, Databases of individual protein families, 3iUD, AdPriBoDB, Animal Toxin Database, ARAMEMNON, BACTIBASE, BYkdb, ChromDB, CLIPZ, ComSeServer, cpnDB, CREMOFAST, CSDBase - Cold Shock Domain database, Cyanolyse, CyMoBase, DanCer, Deam domain database, Defensins Knowledgebase, Degradome Database, DEODb, DOBISCUIT, DSD, EENDB, ERPO-Moscow, FCP, Gene3D, GPCR NaVa database, GPCRPdb, gpDB - G-protein database, Histone Protein Database, Histome, HIV RT and Protease Sequence Database, JIUVC, KiDiFamMap, Kinomer, KLIFS, KnotProt, Knottin database, Laminin Database, Lipase Engineering Database, LIMF MAPS, LOX-DB, MethaCys, MiPMoDB, MoonProt, NORINE, NuclearDB, NURSA, Oeath - now available at RhesusBase, Olfactory Receptor Database, OMPDB, Peptaibol, PeroxiBase, PHYTOPROT, PLANT-pl, PlantsPlants, PLantTribes, PLPMDb, Polbase, PREX, ProGlycProt, ProPeptide, Protein kinase resource, PygRT, RatProt, REBASE, Ribonuclease P Database, RmRdb, RPG - Ribosomal Protein Gene database, SDAP, SenelODB, SCENT, SCVID, SRPDB, SuperCyp, Telomerase database, Thyme, TopFiND, TransportDB, TRIP, UCSD-Nature Signaling Gateway Molecule Pages, UUUC, VKCDB - Voltage-gated K+ Channel Database, WDSPPdb, WERAM, Wnt Database, ZIFDB, Structure Databases, AMDB, ARTS-DB, BacDice, COVID19db, CyanoOmicsDB, EnsEmbl COVID-19 resource, ES, GMRepo, GTDB, gutMGene, KNOTTIN, LPNS and TYGS, mBodyMap, MVIP, NMDC Data Portal, OMIA, PHI-base, PencilHB, SCOVA-MD, Sevens, SubtiWiki, T-cell COVID-19 Atlas, VarEPs, VexPathDB, VFDB, VIMB, ViRBase, VTUnther, ZOEVER, Transcriptional regulator sites and transcription factors, 4AatLAS, Small molecules, BitterDB, CHEBI - Chemical Entities of Biological Interest, ChemBank, ChemDB, Crystallography Open Database, CSD - Cambridge Structural Database, EDULUIS (currently unavailable), FragmentStore, Hemolytic, Het-PDB Navi, HiC-Up, MGSLIN, mVOCC, PDB-Ligand, PubChem, RE.D.B., SCRIPDB, SuperDrug, SuperScent, SuperToxic, Carbohydrates, Bacteroid CSDB (BCSDB), Carbohydrate Structure Database (CarbBank), CCS - Carbohydrate Structure Suite, Glycan, Glycan Fragment DB, Glycoconjugate Data Bank, GlycomapsDB, GlycomeDB, GlytoCan, PCFSDB, SuperSweet, SWEET-DB, UniCarbKb, Nucleic acid structure, 3DNAalNucleic, BIGNASim, Gmsind, GRSDb, IntAct, IS2, MetaPDB, MINAS, NCIR - Non-Canonical Interactions in RNA, NDB, NNDB, non-B DB, PseudoBase+, RNA Bricks, RNA CoSMMs, RNA FRABase, RNA STRAND, RNALunction, SARS-CoV RNA SSS, SMMRNA, Vir-Mir db, VoronoiaRNA, Protein structure, 2P21db, 3DID - 3D interacting domains, ApoHolodB, ArchDB, AS-ALPS, AutoPSI, BioMagResBank, BISc, Bisc, BSDB, CADB - Conformational Angles Database of Proteins, CAPS-DB, CATh, CC-, cPDB, ComSin, ConSurf-DB, CPDB, CUBE-DB, D2P2, DALI database, DARIC, DBAIL, Decors-R-Ujs, DisProt - Database of Protein Disorder, Dockground, DomIns - Database of Molecular Insertions, DOMMINO, DSDBASE, DSM - A database of Simulated Molecular Motions, ECOD, eSite - Electrostatic surface of Functional site, ELM, EMDataBank, FPDP, FuzDB, Gene3D, Genomic Threading Database, GTOP - Genomes To Proteins structures, HOMSTRAD - Homologous Structure Alignment Database, HotSpot, IDEAL, IEDB-3D, iGNM, IMGT/3Dstructure-DB, ImotyR, Jali, Jenali: Jane Library of Biological Macromolecules, KinkDB, LinkProt, LPFC, MALISAM, MegaMotifBase, MMDB, ModBase, MolMovDB - Database of Macromolecular Movements, NPIDb (Nucleic acids ÅĤĀĈĀĜĀĜĀ-ĀĈĀ, "Protein Interaction Database), NRC-Clinc, OPN, PASS2, PCDB, PCDDb, PDB facilities, PDB-REPRD, PDBflex, PDB, PDBselect, PDBsum, PDB_TM, pE-DB, PFD - Protein Folding Database, Phospho3D, PIDDT, piSite, Platinum, PMDB - Protein Model Database, Pocketome, PoSuM, ProSAs, ProTChemSI, Protein Geometry Database, PRTAD, PSCDB, PSI-SCKB, PTCL, RCSB Protein Data Bank, RepeatsDB, RESID, Rstadb, SAHG, SASDB, SCOP2 - Structural Classification Of Proteins, SCOPE, SCOPPI, SIFTS, SISYPHUS, SMOs2: Update of database of Structural Motifs in Superfamilies, SNAPPI, SPROUTS, SSToSS - Sequence-Structural Templates of Single-member Superfamilies, STAP refgen, SomaMutDB, StartFold, STINGreport, Structure Superposition Database, SUPERFAMILY, SWISS-MODEL Repository, TargetDB, TOPSAN, TSTM, ValidatorDB, VIPERdb, Voronoia, WALTZ-DB, Genomics Databases (non-vertebrate), Animal QTldb and CorrDB, Compodynamics, EchinoBase, EGA, EWAS Open Platform, FlyAtlas 2, Gene Expression Nebulas, Genomicus, HCD, InsectBase, iSYTE, MetaExp, miTED, PhylomeDB, PopHumanVar, proCHiPB, Regeneration Roadmap, rgdm, SomaMutDB, SPICA, webTWAS, Genome annotation terms, ontologies and nomenclature, BioGPS, BioThesaurus, BTO, dcGO

Yes, you can
keep zooming
in :)

GENE BANK

The NCBI GenBank

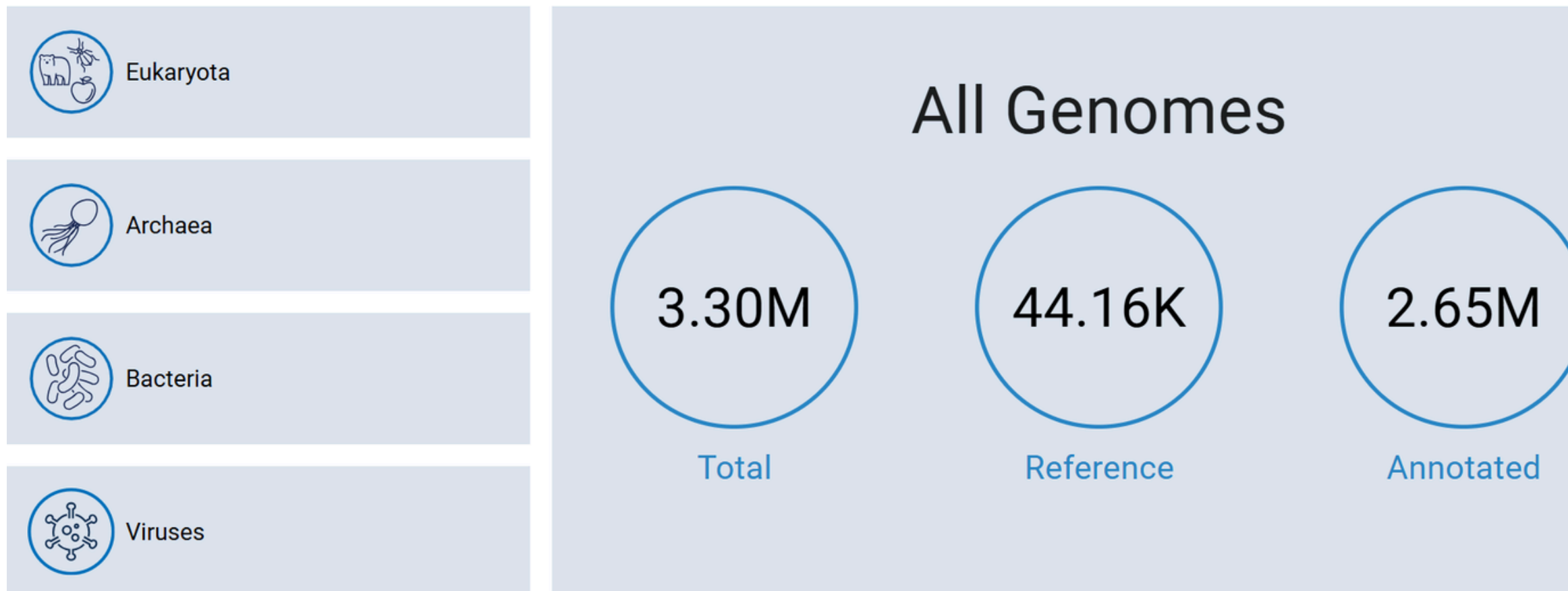


National
Center for
Biotechnology
Information



<https://www.ncbi.nlm.nih.gov/genbank/>

The NCBI GenBank



As a first example, try typing in “homo sapiens”

Search term

Try examples: [Homo sapiens](#) [GCF_000001405.40](#) [PRJNA489243](#) [SAMN15960293](#) [WFKY01](#) [GRCh38.p14](#) [NC_000913.3](#)

The NCBI GenBank

Take a glimpse at our own genome first, then download the reference genome!



View annotated genes



See in Genome Data Viewer



BLAST the reference genome



Compare genomes

Reference genome

[GRCh38.p14](#)

Genome Reference Consortium (2022).

RefSeq GCF_000001405.40

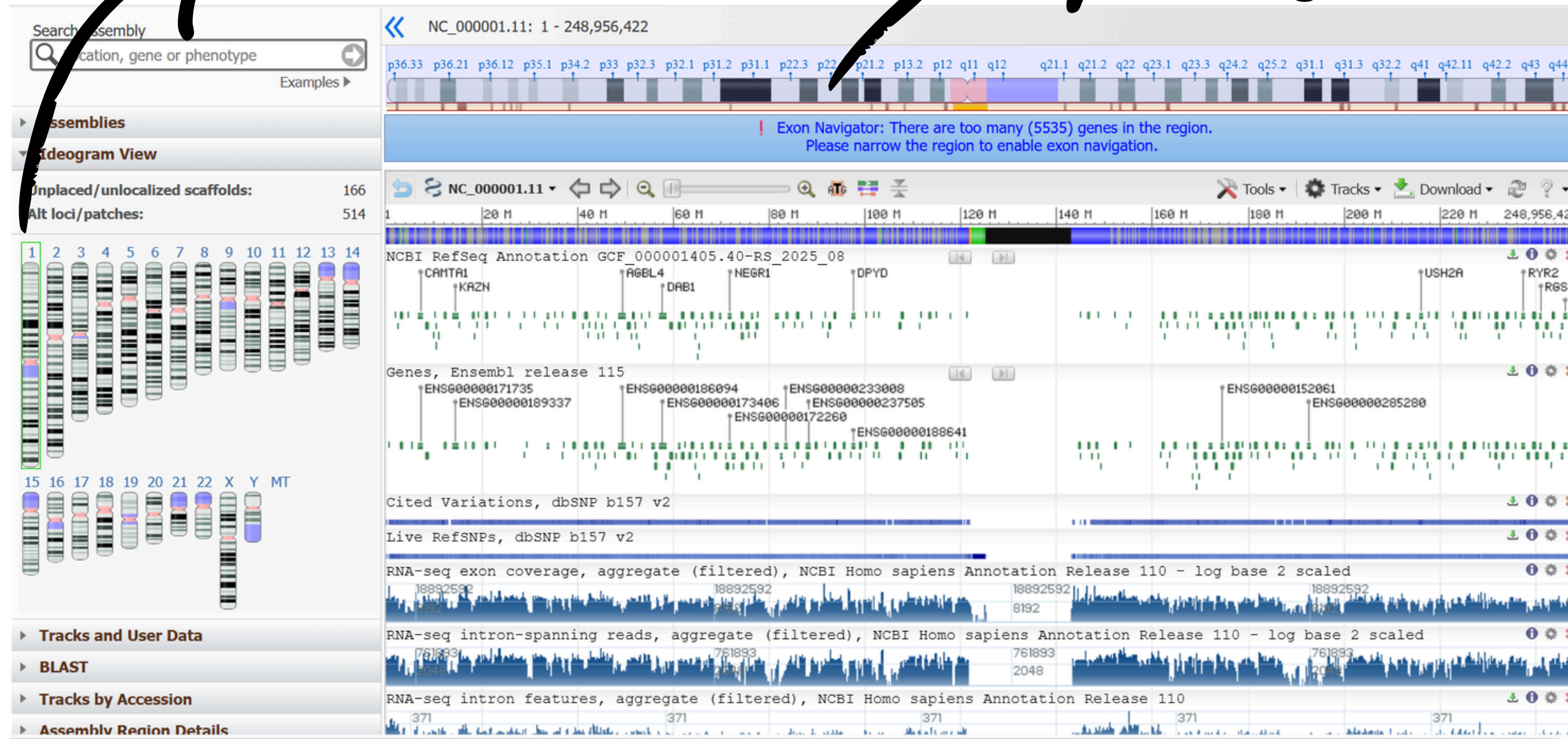
Download

Some cool stuff



Change the chromosome view

Zoom in a bit, explore the exon navigator



- Search for ACSS2 in the search assembly bar. What can you find?

Homo sapiens
(human)

Search assembly

ACSS2

Examples ▶

Some cool stuff



1. Select a species

Homo sapiens (human)



3. Select an assembly

GRCh38.p14 (GCF_000001405.40)



Clear Form

View Comparison

2. Select a second species

Mus musculus (house mouse)



4. Select a second assembly

GRCm38.p6 (GCF_000001635.26) C57BL/6J



How alike are we?

UNIPROT

Universal Protein



<https://www.uniprot.org/>

Universal Protein

| **Function**

Names & Taxonomy

Subcellular Location

Disease & Variants

PTM/Processing

Expression

Interaction

Structure

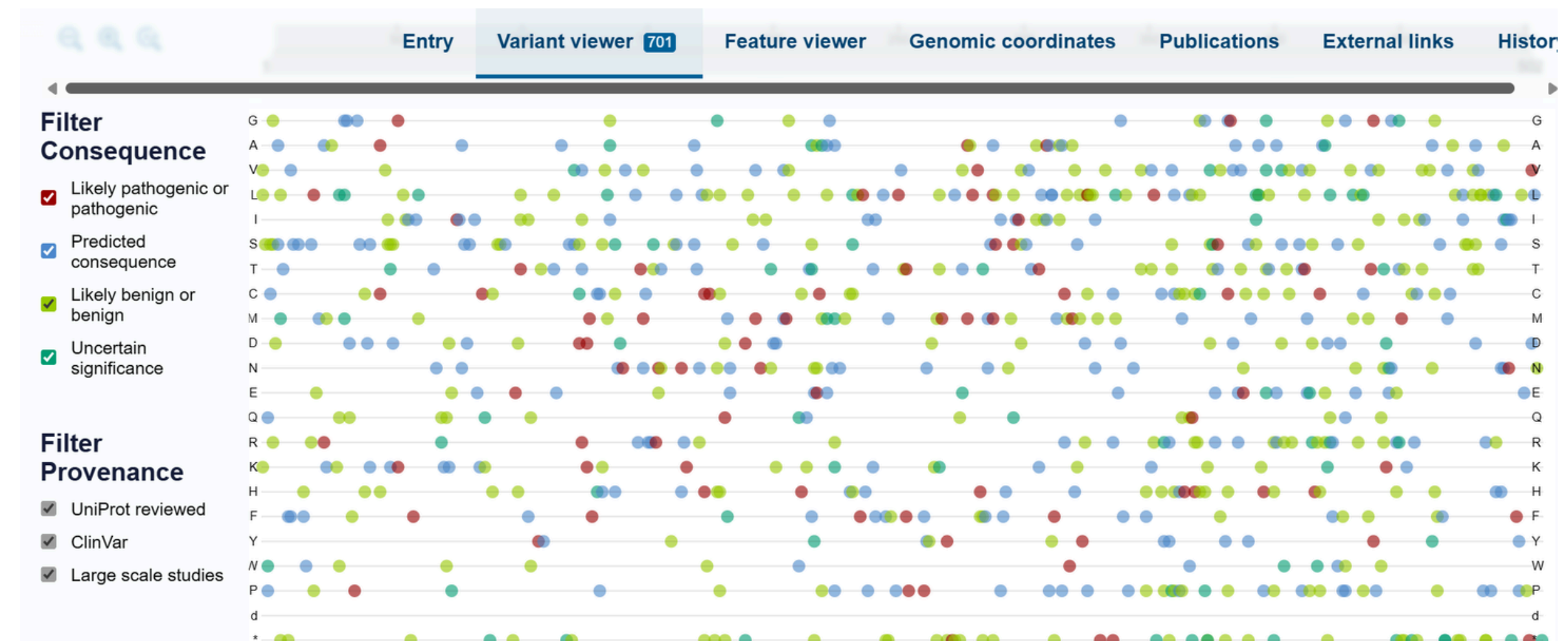
Family & Domains

Sequence

Similar Proteins

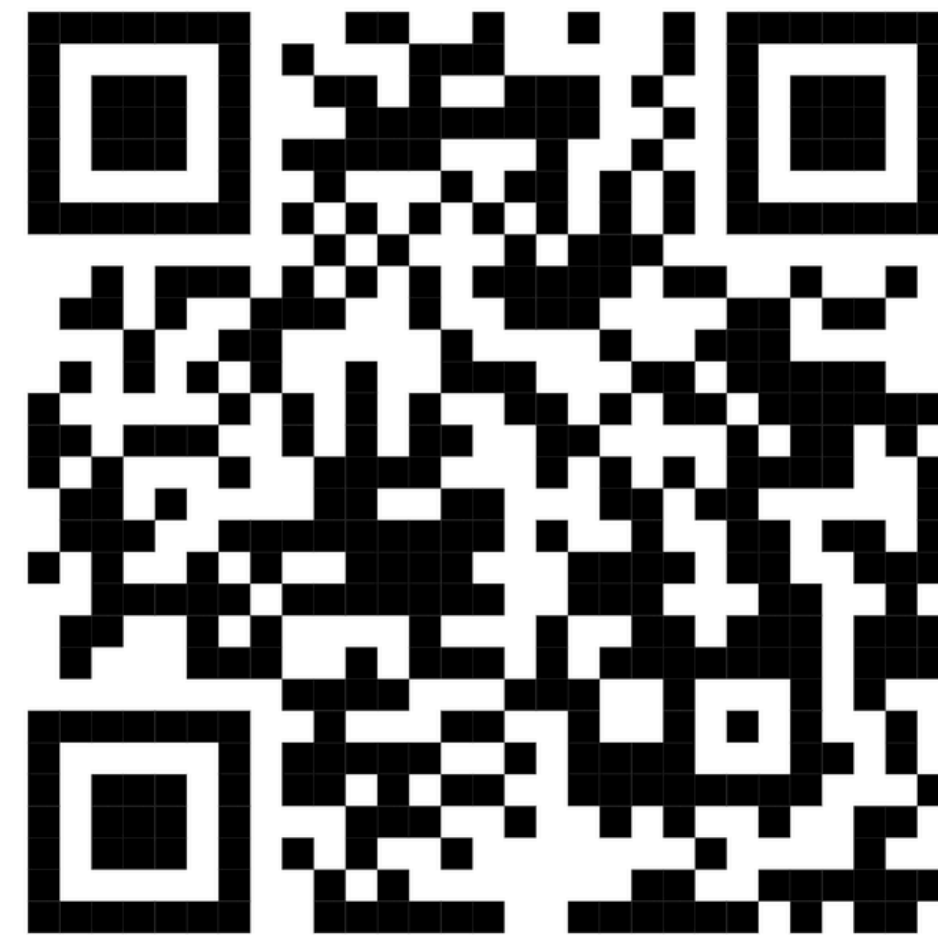
Now that you know what is ACSS2, look for your protein in UniProt

- What information can you find?
- Try to understand the variant viewer, what are you seeing?



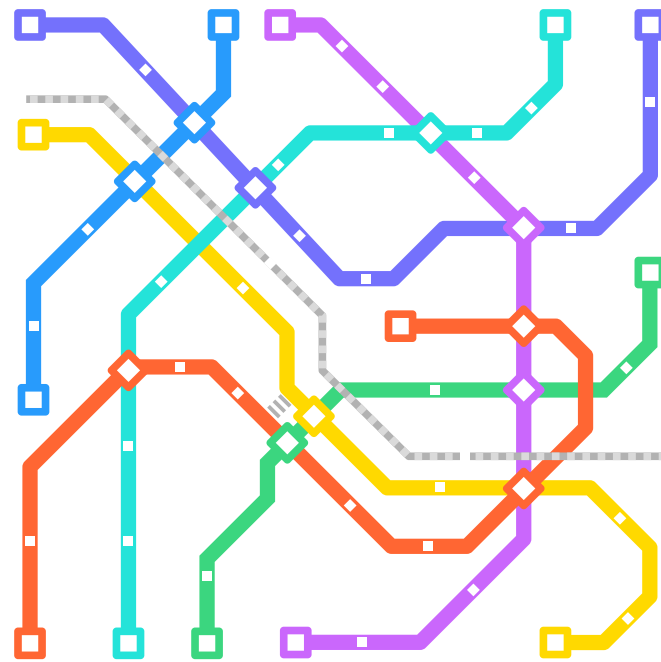
KEGG

Kyoto Encyclopedia of Genes and Genomes



<https://www.genome.jp/kegg/>

Kyoto Encyclopedia of Genes and Genomes



How'd you find the pathway?



KEGG PATHWAY Database

Wiring diagrams of molecular interactions, reactions and relations

[KEGG2](#) [PATHWAY](#) [BRITE](#) [MODULE](#) [KO](#) [GENES](#) [COMPOUND](#) [NETWORK](#) [DISEASE](#) [DRUG](#)

Select prefix

map

Organism

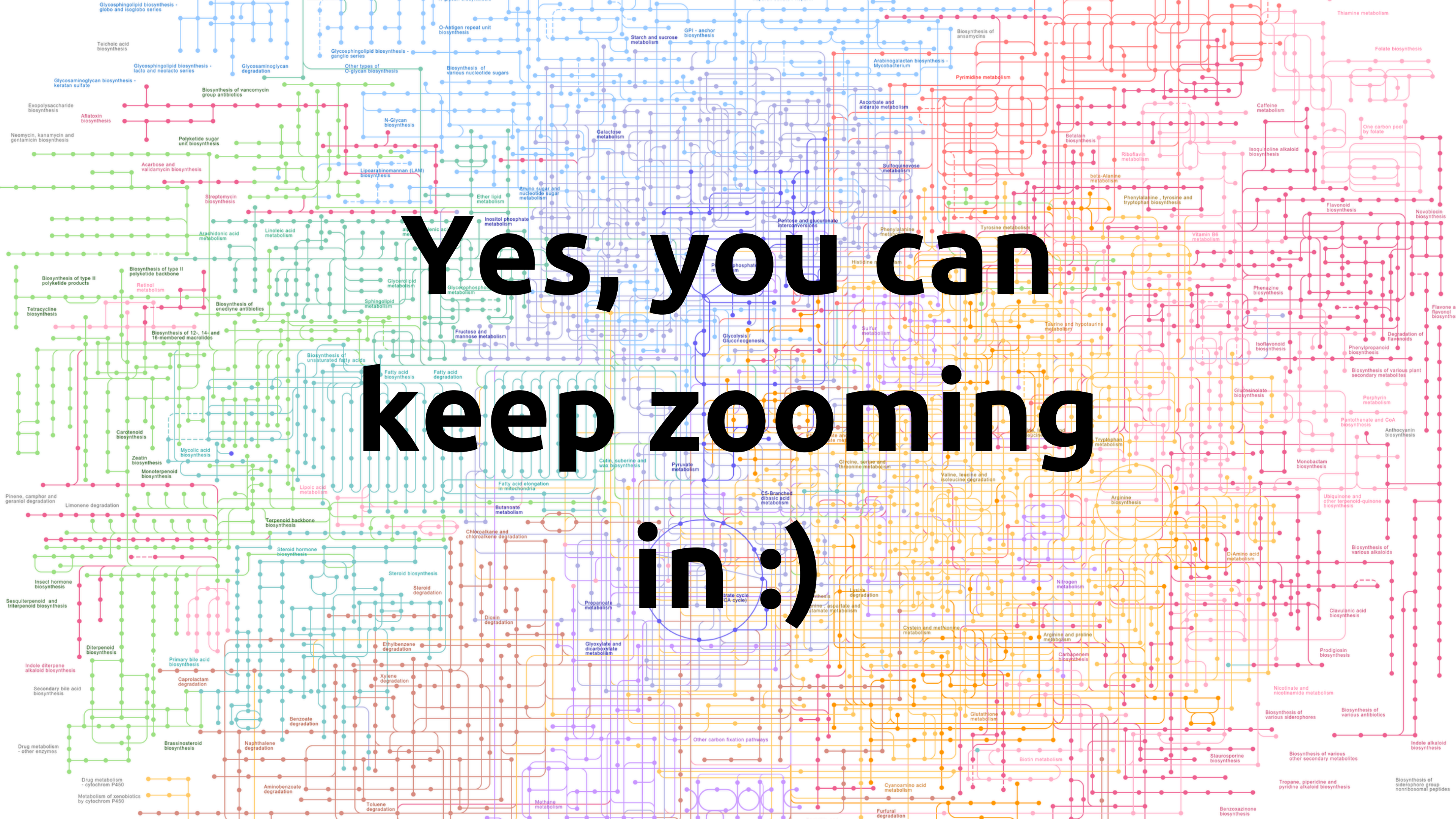
Enter keywords

hsa Acetyl-CoA

Go

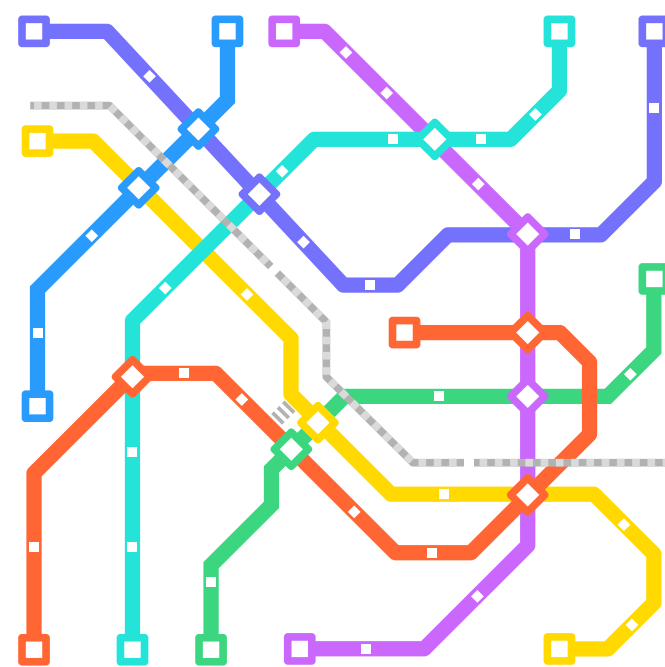
[Help](#)

Try typing in “hsa Acetyl CoA”



Yes, you can
keep zooming
in :)

Kyoto Encyclopedia of Genes and Genomes



Now just try typing in map00020



KEGG PATHWAY Database
Wiring diagrams of molecular interactions, reactions and relations

KEGG2 PATHWAY BRITE MODULE KO GENES COMPOUND NETWORK DISEASE DRUG

Select prefix Organism [Help](#)

What do you think was the main difference between the two search entries?

Thanks !!!!!